Izma Fatima Mohammad

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Pakistan

Education

Bachelor of Engineering, Biomedical Engineering| **CGPA**: 3.82/4.00 | 3rd year (6th Semester)
Ziauddin University | 2024 – present (*expected December 2027*)
BCI & Neurotechnology Spring School 2026 | April 2026 | Awarded Hours: 130
By **g.tec Medical Engineering, Austria**

Skills

Computer: MS Office, Adobe Suite, Arduino IDE, PASCO Capstone, Kinovea, MultiSim, NI Elvis, Materials Studio, SOLIDWORKS, VS Code, Godot, Unity, MATLAB, COMSOL Multiphysics, Python, C#, C, Lovable.dev, Prompt Engineering, Notion, Figma, FlutterFlow.
Language: Fluent in English and Urdu; conversational skills in German, Korean, and Mandarin.
Leadership: Strategic product vision and ownership, multidisciplinary team coordination, product strategy and systems thinking, problem discovery and definition, initiative, autonomy, and end-to-end ownership, technical translation and stakeholder alignment, resource-conscious prioritization and decision-making.

Projects

- **XAI-STEM: An Explainable AI Framework for Personalized and Teacher-Assisted STEM Education**
  - *August 2026 (Ongoing) | Team Member (Product Progression Director)*
  - Developing a human-centered Explainable AI framework that assists STEM teachers in identifying learner needs, personalizing instruction, and making data-informed teaching decisions through transparent and interpretable AI recommendations.
- **Personalized Adaptive Exam Intelligence System (PAEIS): An AI-Driven Framework for Continuous Learner Modeling and Adaptive Exam Preparation**
  - *June 2026 (Ongoing) | Individual Project*
  - PAEIS is an adaptive artificial intelligence framework that continuously models a student's academic state by analyzing their responses, errors, syllabus-level mastery, command-word performance, retention, and examination behavior. It integrates this learner model with curriculum and assessment knowledge to identify knowledge gaps, diagnose recurring misconceptions, predict performance, and dynamically select the most appropriate learning intervention, question, revision activity, or feedback at each stage of preparation
- **SIM-IE: Biomedical Engineering Virtual Simulation & Digital Twin Learning Platform**
  - *April 2026 (Ongoing) | Founder & Lead Developer*
  - Developing a web-based biomedical engineering simulation ecosystem focused on project-based and problem-based learning through interactive virtual laboratories, biomedical equipment simulations, and digital twin environments.
  - Designing educational and technical modules integrating biomedical instrumentation, service engineering training, simulation-driven experimentation, and collaborative learning systems for biomedical engineering students and trainees.
  - Integrating simulation technologies including Unity and web technologies (HTML, CSS, React) to create scalable high-fidelity and low-fidelity biomedical simulations.
  - Conceptualizing future expansion toward VR-based immersive simulations, OEM-integrated device training, and global collaborative learning infrastructure aligned with Washington Accord CLOs and PLOs.
- **Developing a 2D skin model that demonstrates the effect of skin roughness on melanoma detection during IR Thermography**
  - *January 2026 (ongoing) | Mentee*
  - Made using COMSOL MultiPhysics. Modelling done based on Pennes bioheat equation and the rough surface by using a double-sum parametric equation that synthesizes a truncated Fourier-like series of elementary cosine waves modulated by a Gaussian random amplitude attenuating via a spectral exponent and randomized by uniform phase shifts.
- **Code Blue, 3D game development project using Unity**
  - *January 2026 (ongoing) | Team Lead*
  - Developing a simulation game that allows biomedical engineering students to learn about biomedical equipment in virtual simulated environments while experiencing the job of a service engineer in the radiology department of a hospital in apocalyptic settings.
- **3D Skin Capillary Model for More Accurate Simulative Studies**
  - *January 2026 (ongoing) |Individual Project*
  - Developing a 3D skin capillary model in MATLAB for an advanced skin model for a more realistic model to improve simulative diagnosis for skin cancer and melanoma detection.
- **Presentation of Virtual Simulation Platform for Biomedical Engineering Education on IBFT'26 (SSUET & UTM)**
  - *January 2026 | Solo Project*
  - Prepared a presentation on one of my ongoing personal projects on the Virtual Simulation Platform for Biomedical Engineering Educational purposes, which allows collaborative project- and problem-based learning.
  - Presented among senior researchers from Pakistan, Malaysia, the United States, and Italy.
- **Connector.global: An Educational Ecosystem**
  - *November 2025 (Ongoing) | Project Director*
  - Ideated, planned, and implemented the project to make quality education accessible to everyone to *Reduce Inequalities* via *Partnerships* with experts in specific domains in education. Working exclusively on SDG 4, 10, and 17. Bringing research-backed programs to provide students with Global Opportunities. As a result, creating global impact in terms of technology, education, collaboration, global citizenship, and sustainable development.
  - Worked with juniors, seniors, experts, and freshers to achieve set goals.

- **Analysis of Postural and Gait Deviations During Bicep Curl Exercise Among Gym Trainees**
  - *September – December 2025 | Team Member*
  - Conducted a statistical Analysis on the current performance of the bicep curl exercise in gym trainees across Karachi, Pakistan.
  - Analyzed the bicep curl performance using the software *Kinovea*
  - Concluded that many trainees have recurring errors in their posture while performing the exercise, leading to adverse gait deviations and chronic pain in the neck and lumbar region.
- **Basic hearing aid circuit design for sound amplification & noise reduction**
  - *April – May 2025 | Team Lead*
  - Applied the understanding of amplifiers, filters, sound signals, and the anatomy of the ear to create an economic circuit design that effectively amplifies the sound and reduces noise in sound signals.
  - This project allows our team to fully grasp future project advancements in creating a cost-effective device.
- **4 Digit Electronic Code Lock System**
  - *August – December 2025 | Team Member*
  - A cost-effective digital lock system made using basic logic gates and electronic components. The project focuses on applying digital logic concepts in a practical, real-world application and provides hands-on experience in circuit design, assembly, and testing
- **UI & UX for Gaba Education website**
  - *June – August 2025 | Main Director*
  - The website the organization already had was outdated and had multiple crucial UI/UX design fails. Me along with my team of 2 people worked with the GABA Education to perfect their user's experience of their website.
- **Cost-Effective Mini 4 Propellers Drone using ESP32 and MPU6050**
  - *January – May 2025 | Team Member*
  - 4 propellers 3D printed drone made using ESP32 microcontroller and MPU6050 sensor over the course of 5 months.

Experience

- **Private Mentor**| 2026 – present
  Leading students with a unique mentorship package of curricular studies tutoring along with venture projects mentorship, scientific research mentorship and global professional networking guidance. Helping students in setting new levels of achievements each academic year.
- **Research Intern** at UTM-IJN, Malaysia | July 2026
  Worked under Prof. Madya Ir. Dr. Tan Tian Swee, Associate Professor in the Department of Biomedical Engineering and Health Sciences at Universiti Teknologi Malaysia (UTM). Working on a research project regarding, XAI for diagnostics of Oral Cancer.
- **Intern** at JPMC, Department of CyberKnife and Tomotherapy | June 2026
  Understood the working principles, operations, and treatment procedures in CyberKnife S7, CyberKnife VSI, and Tomotherapy X9 by Accuray.
  Closely observed error occurrence, troubleshooting methodologies and error handling with service engineers.
  Expansive study on medical devices in Robotic surgery, ICU, HDU, ENT OT, PET, Radiology, Diagnostic Laboratories, Blood Bank and Neurotrauma departments.
- **Project Director** at Genesis Communications, Connector.pk | 2025 – Present
  Working with the Technology Integration Team, Marketing Team, Outreach Team, Partnership Team, and Academics Team.
- **Private Tutor** | 2022 – 2025
  Taught various grades, multiple subjects. Subjects including Mathematics, Computer Science, Biology, Chemistry, Physics, Islamic Studies, Urdu. For grades 5 till A levels. Proven results of 20% grade increment in school and examination boards tests.
  Successfully taught a 13- year old from O levels to A levels as she passed her CAIEs at the age of 13.
- **Book Developer** at Genesis Communications | 2023 – 2025
  Performed duties by drafting Content Lists for textbooks for subjects, i.e. Science, Mathematics, Computer, and English, that aligned with *SNC (Single National Curriculum of Pakistan).* Worked in collaboration with subject specialists to write chapters for textbooks for grades 1-8. Worked with Graphic Designers to create initial book layouts.

Co-curricular Activities

- **Director of Public Relations**, IEEE Ziauddin University | 2026 - Present
- **General Secretary**, Ziauddin University Student Council | 2026- Present
- **Graphic Designer** (Marketing Team), IEEE Ziauddin University Student Branch | 2024 – Present
- **Volunteer**, Dar ul Sukoon | 2025
- **Volunteer**, Ziauddin University Student Council | 2024

Competitions & Hackathons

- **BR4IN.io by g.tec Medical Engineering Austria** | 2026
  Participated in Brain-Computer Interface Hackathon 2026. Project: P300 Data Analysis for a P300 Speller Data was collected from five healthy subjects (S1–S5) using the Unicorn Brain Interface (8 channels, 250 Hz). Worked with a group of professionals from Italy and France.
- **Pro Battle'25 by IBA Karachi** | 2025
  Built a fully functional LFR (Line Following Robot) using servo motors, Arduino UNO, and a LiPo battery.

Certifications

- **Regenerative Medicine and Stem Cell Conference (RSC – 2026)**
  By Accreditation Council for Continuing Medical Education (ACCME)

- **Declamation Contest: Life and Contributions of Quaid-e-Azam M. Ali Jinnah**| 2026
  By Ziauddin University
- **Materials: Modeling and Simulations Using Materials Studio**| 2024
  By BME Dept. SSUET, IEEE EMBS SSUET, BIOMED 5.0 Project
- **Use Care & Maintenance of Scientific & Laboratory Instruments**| 2024
  By Government of Pakistan, Ministry of Science & Technology, PCSIR Laboratories Complex